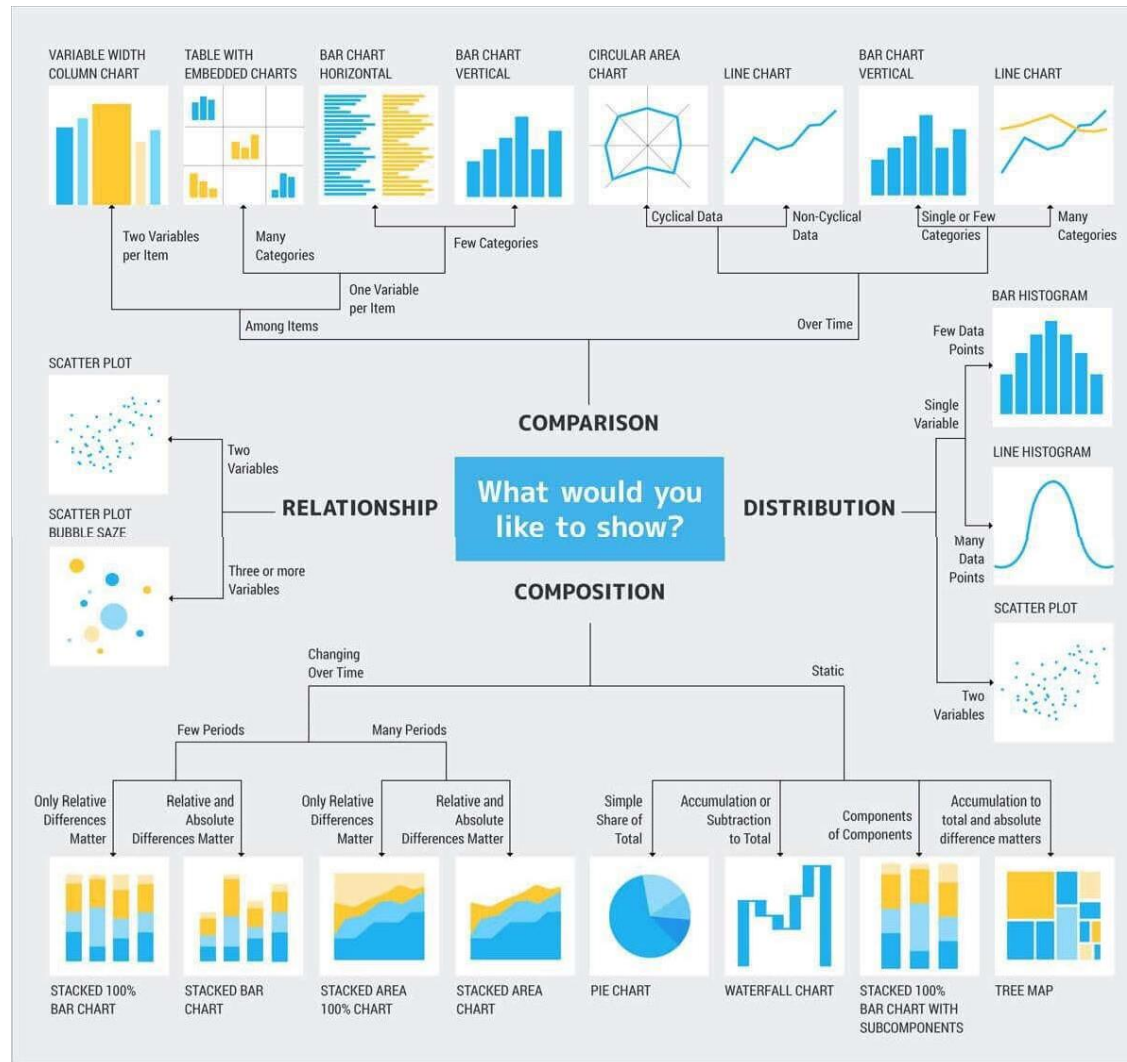


Introduction to Data Science



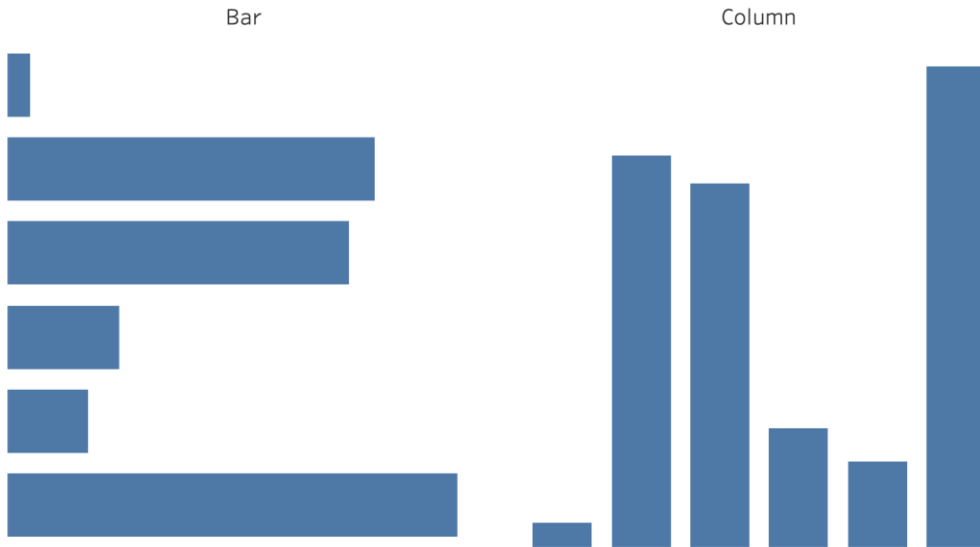
Dashboards and Storytelling

Do I have the right chart?

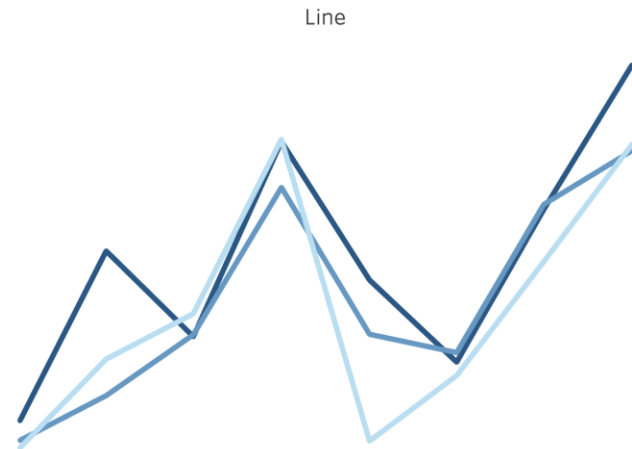


Comparison

Among Items



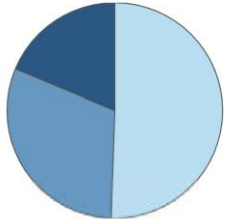
Over Time



Composition

Static

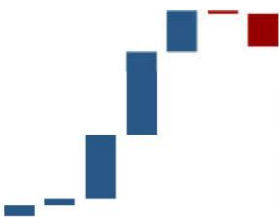
Pie



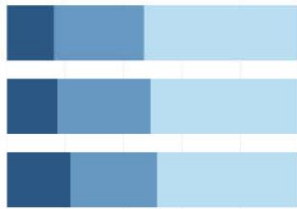
Tree



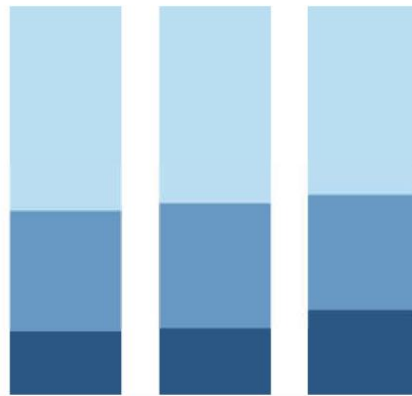
Waterfall



Stacked Bar

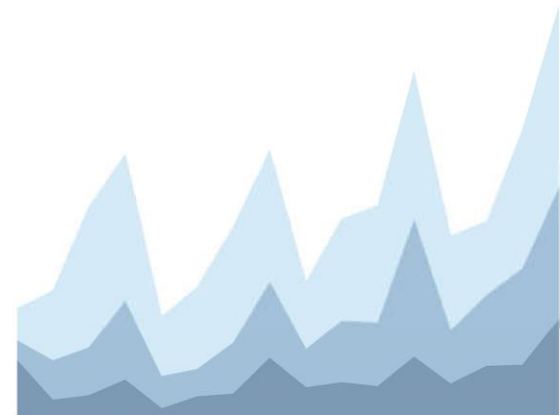


Column Stacked



Over Time

Stacked Area

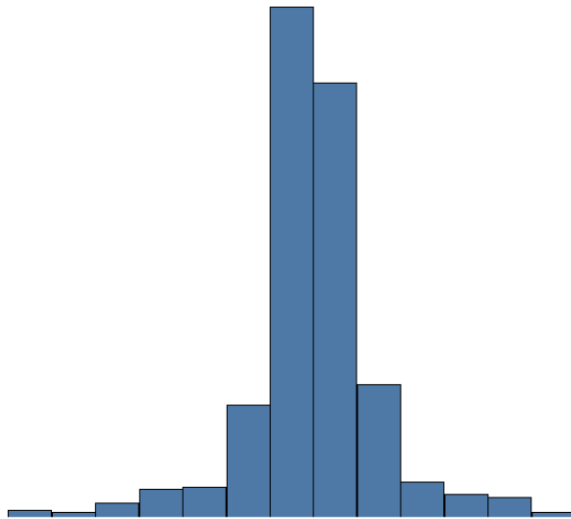


Distribution

Single Variable

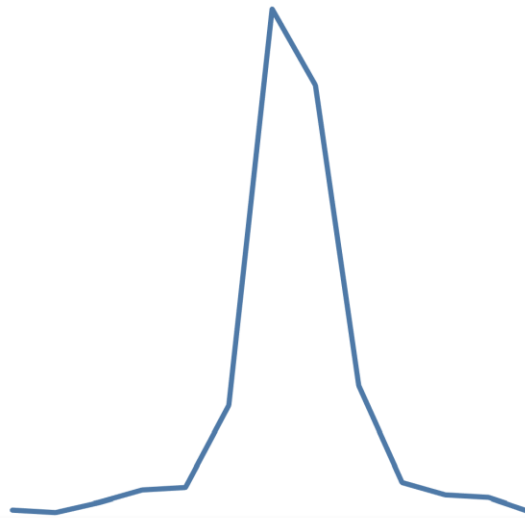
Few Data Points

Column Histogram



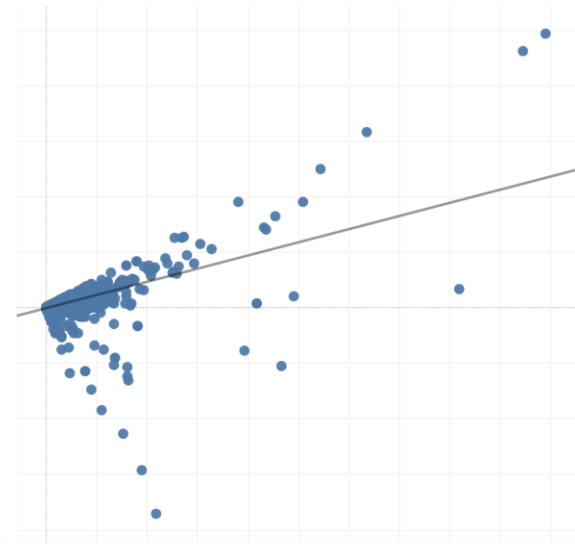
Many Data Points

Line Histogram



Two Variables

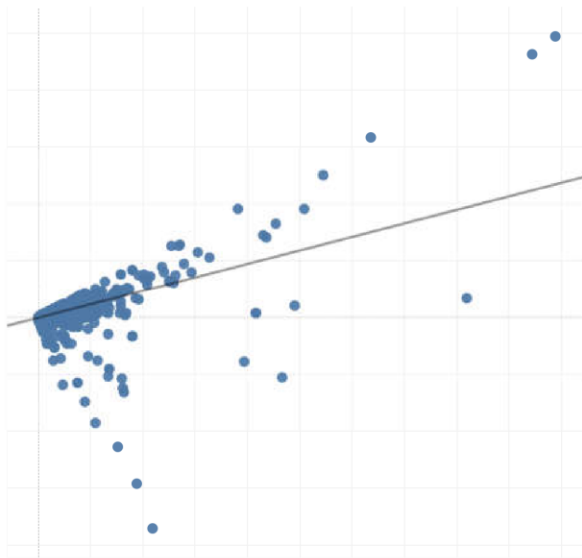
Scatter



Relationship

Two Variables

Scatter



Three+ Variables

Bubble



Dashboard Flow

What is a dashboard?

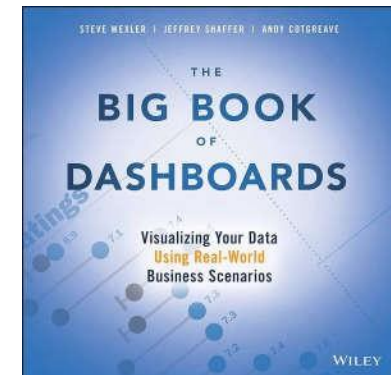
“A dashboard is a visual display of the most important information needed to achieve one or more objectives; consolidated and arranged on a single screen so the information can be monitored at a glance.”



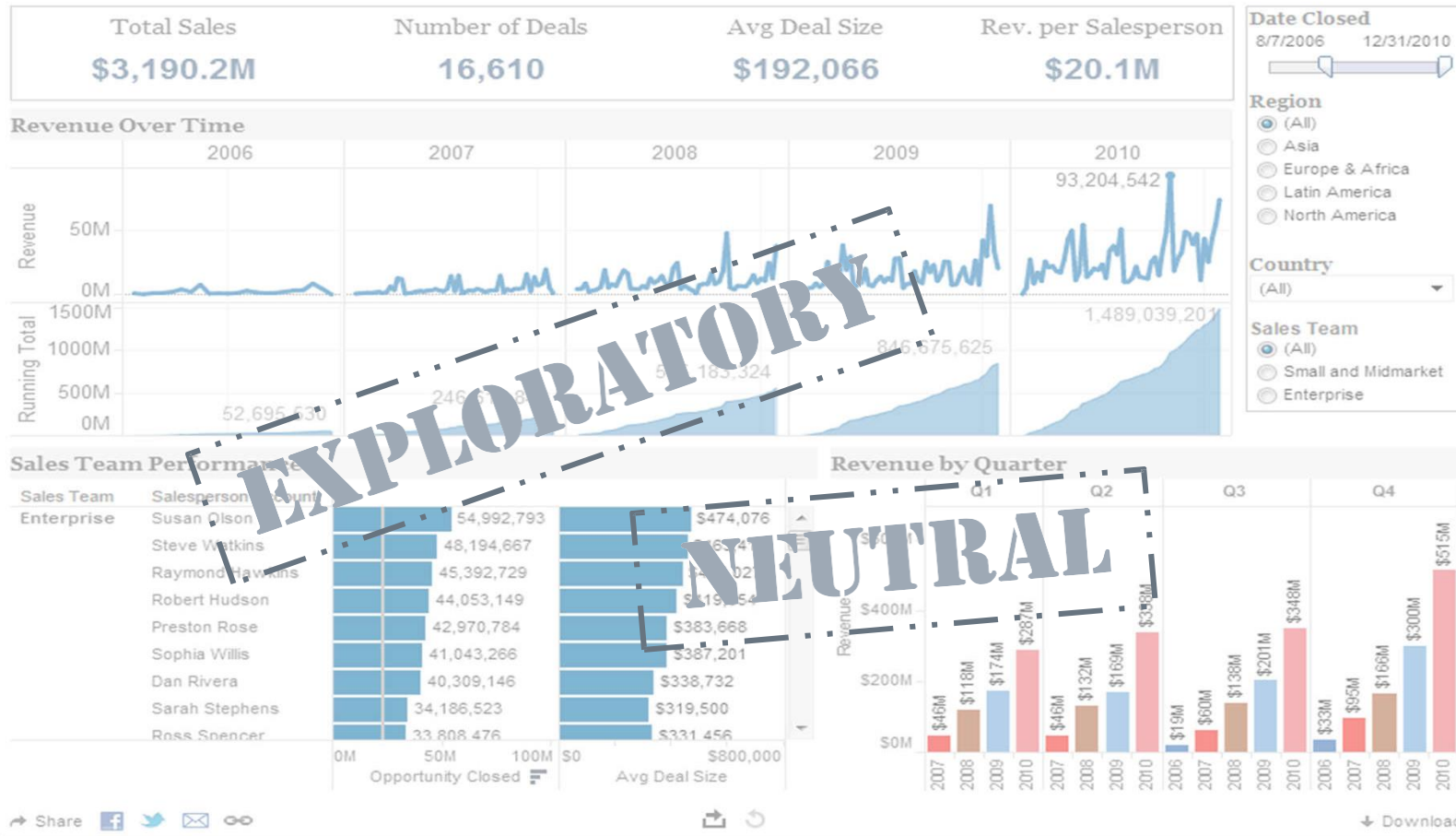
*Stephen
Few (2004)*

“A dashboard is a visual display of data used to monitor conditions and/or facilitate **understanding**”

*Big Book of
Dashboards
(2017)*



Sales Dashboard



PHILADELPHIA CRIME SCENE

2006
92,855
incidents

OVERALL CRIME DROP IN
2013 BY 20%
COMPARED TO 2006

In, 2008 there were **90,644** incidents
Dropping crimes down by **2.4%** compared to 2006

In, 2011 there were **83,225** incidents
Dropping crimes down by **10.4%** compared to 2006

In, 2013 there were **74,592** incidents
Dropping crimes down by **19.7%** compared to 2006

EXPLANATORY

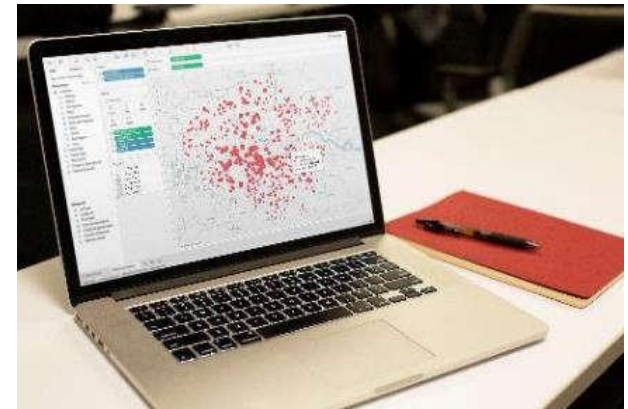
OPINIONATED

HOVER OVER
THE LINE CHART

TO SEE WHICH CRIME TYPES
OCCURED IN PHILADELPHIA OVER
TIME

What makes a good dashboard?

- Answers a set of questions
- Follows a flow and invites interactivity
- Condensed; primarily in the form of summaries and exceptions
- Specific to and customized for the dashboard's audience and objectives
- Provides appropriate text for clarity and direction, if needed
- Makes strategic use of color



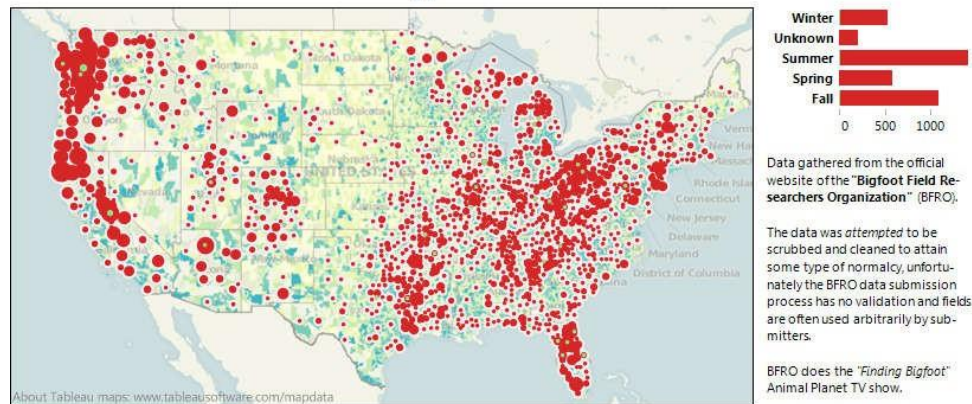
Layout



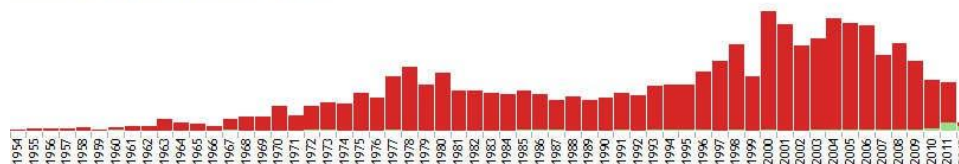
www.useit.com

Layout Top Left

Where is bigfoot seen in the US?



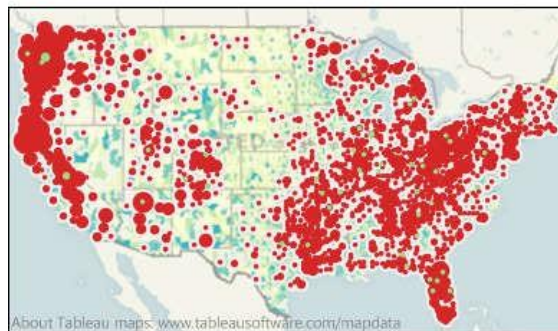
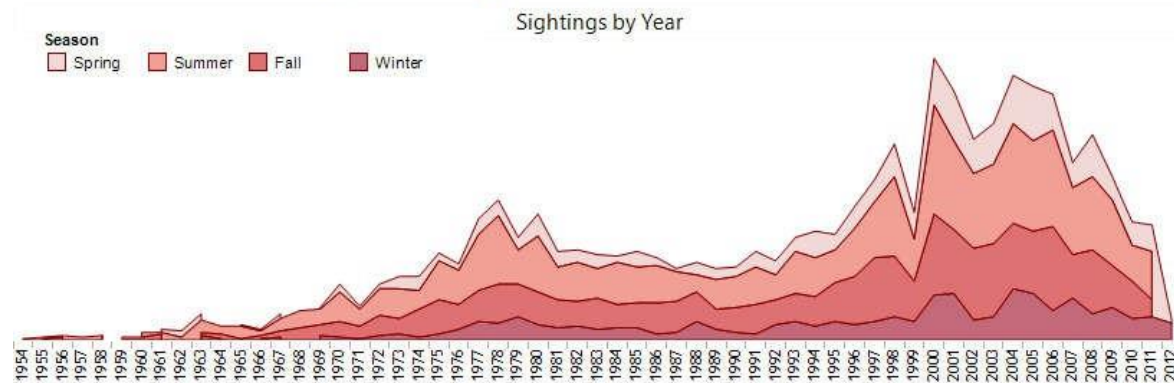
Click on ANY element of the visualization (location, season, year, detail field) in order to filter by that item.
Select the element AGAIN to go back to the full view.



The BFRO classifies sightings according to a system based on the sightings "potential for misinterpretation".

Total Sightings			Class A		Class B		Class C		Unclassified		
3,806			1,951		1,696		31		128		
Alabama	Baldwin County	1979	September	Class A	Man recalls a sighting after Hurricane Frederic north of Mobile						
	Barbour County	1980	January	Class A	Motorists pulled over on a rural highway at night describe standoff in headlights						

Bigfoot sightings are in decline



Data gathered from the official website of the "Bigfoot Field Researchers Organization" (BFRO).

The data was *attempted* to be scrubbed and cleaned to attain some type of normalcy, unfortunately the BFRO data submission process has no validation and fields are often used arbitrarily by submitters.

BFRO does the "Finding Bigfoot" Animal Planet TV show.

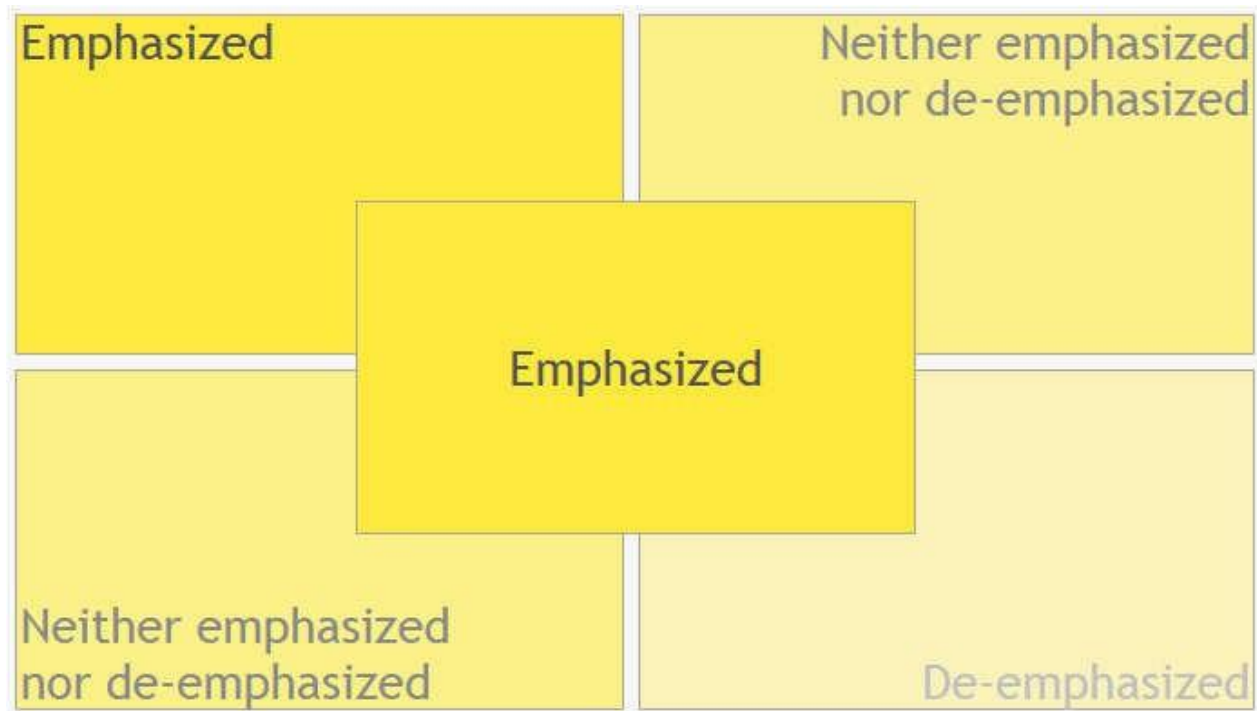
Click on **ANY** element of the visualization (*location, season, year, detail field*) in order to filter by that item.
Select the element **AGAIN** to go back to the full view.

The BFRO classifies sightings according to a system based on the sightings "potential for misinterpretation".

Total Sightings	Class A	Class B	Class C	Unclassified
3,806	1,951	1,696	31	128

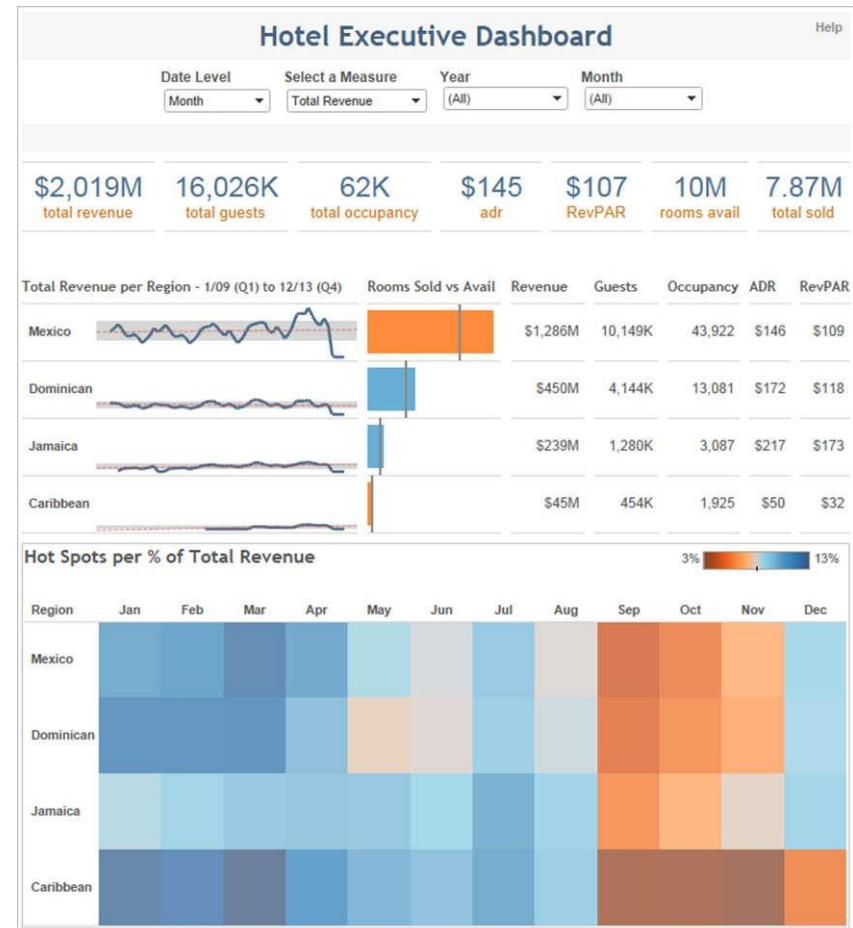
Alabama Baldwin County 1979 September Class A Man recalls a sighting after Hurricane Frederic north of Mobile

What are you emphasizing?



Dashboard Flow

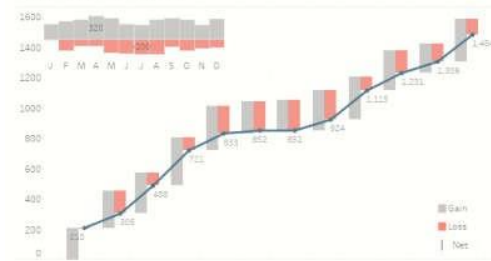
- Filters at the top, so user knows when and how to start interacting
- Most important information at the top and centered
- Starts with the highest level and works down to more precise details
- No more than 4-5 worksheets



Using BANs

Subscriber Churn Analysis

Subscriber activity - All



Net subscriber activity by division



Details

		Gained	Lost	Net	Running total
West	January	80	0	80	80
	February	80	-15	65	145
	March	90	-30	60	205
	April	120	-25	95	300
	May	100	-40	60	360
	June	119	-77	42	392
	July	75	-45	30	422
	August	113	-77	42	464
	September	90	-30	60	524
	October	80	-15	65	589
	November	80	-20	60	649
	December	95	-35	60	709
Central	January	60	0	60	60
	February	85	-45	40	100
	March	80	-27	53	153
	April	90	-17	73	226
	May	120	-33	87	313
	June	45	-30	-35	278
	July	75	-45	30	308
	August	45	-80	-35	273
	September	80	-27	53	326
	October	85	-45	40	366
	November	60	-35	25	391
	December	80	-27	53	444
East	January	70	0	70	70
	February	80	-30	-10	60
	March	100	-30	70	130
	April	110	-45	65	195
	May	70	-25	-25	170
	June	45	-33	-12	158
	July	90	-34	56	212
	August	99	-34	65	277
	September	112	-34	78	355
	October	99	-38	11	366
	November	55	-48	-10	356
	December	110	-45	65	421
Grand Total		1,000	-468	532	532

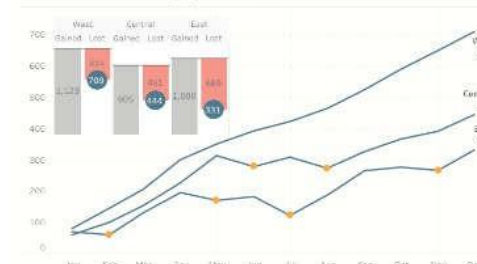
Subscriber Churn Analysis

Gains	Losses	Net	Best month	Worst month
3,028	-1,544	1,484	Apr	Jul

Subscriber activity - All



Net subscriber activity by division



Details

		Gained	Lost	Net	Running total
West	January	80	0	80	80
	February	80	-15	65	145
	March	90	-30	60	205
	April	120	-25	95	300
	May	100	-40	60	360
	June	119	-77	42	392
	July	75	-45	30	422
	August	113	-77	42	464
	September	90	-30	60	524
	October	80	-15	65	589
	November	80	-20	60	649
	December	95	-35	60	709
Central	January	60	0	60	60
	February	85	-45	40	100
	March	80	-27	53	153
	April	90	-17	73	226
	May	120	-33	87	313
	June	45	-30	-35	278
	July	75	-45	30	308
	August	45	-80	-35	273
	September	80	-27	53	326
	October	85	-45	40	366
	November	60	-35	25	391
	December	80	-27	53	444
East	January	70	0	70	70
	February	80	-30	-10	60
	March	100	-30	70	130
	April	110	-45	65	195
	May	70	-25	-25	170
	June	45	-33	-12	158
	July	90	-34	56	212
	August	99	-34	65	277
	September	112	-34	78	355
	October	99	-38	11	366
	November	55	-48	-10	356
	December	110	-45	65	421
Total		1,000	-468	532	532

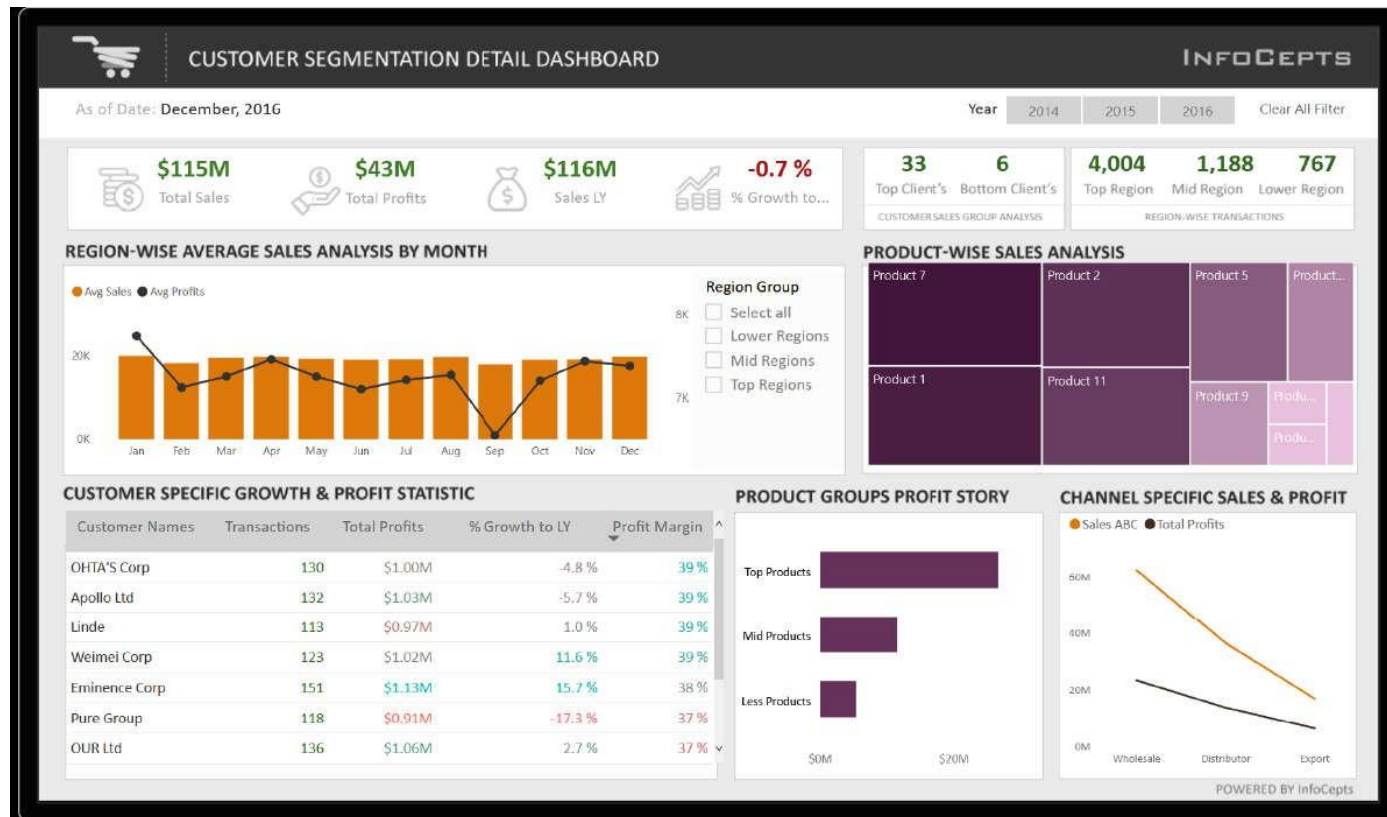
Does the dashboard pass the 5 second test?

- Most important view is on top or top left
- Legends are near their views
- Avoid multiple color schemes
- Uses 5 or fewer views (charts)

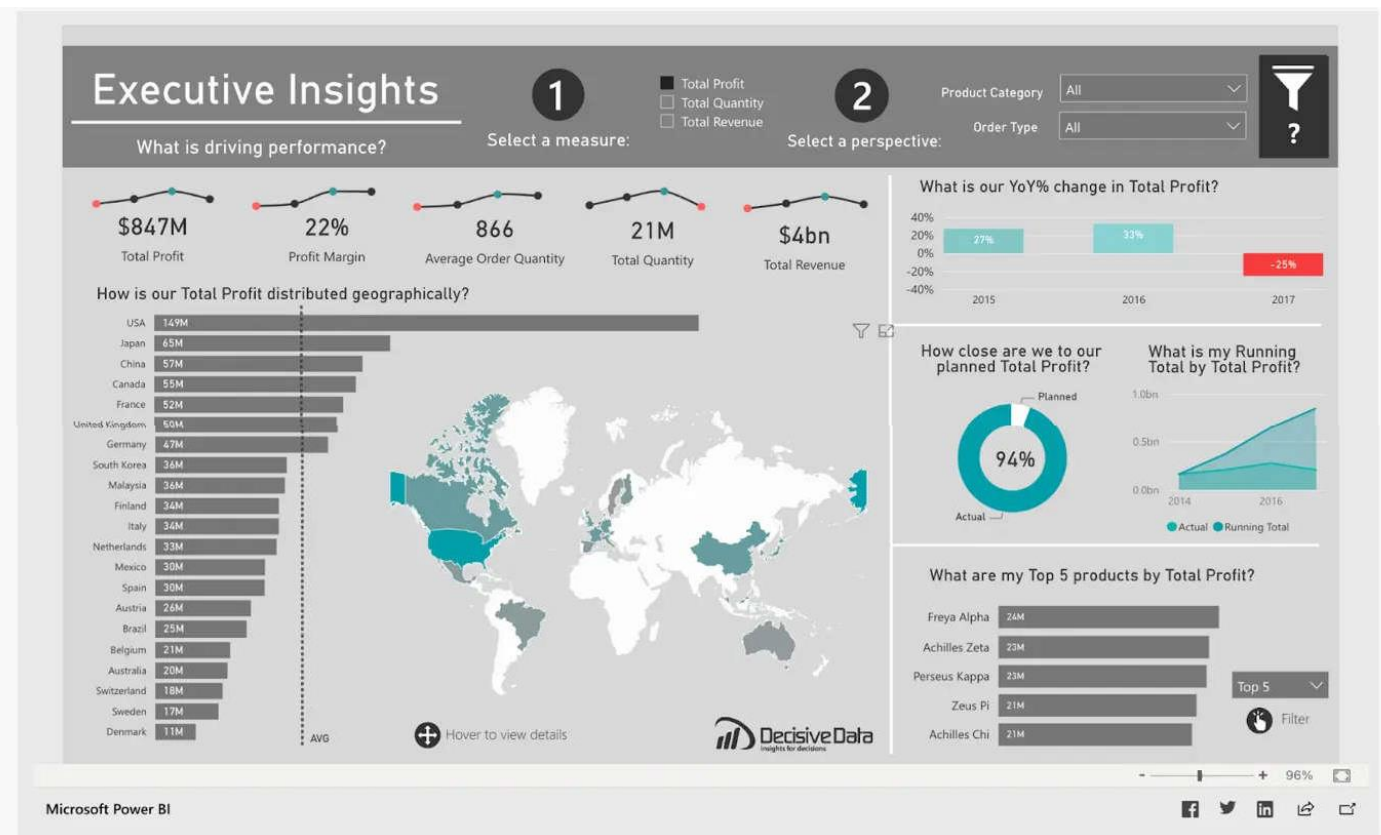
Dashboard Mistakes

- Not creating a clear goal for the dashboard
- Presenting too much information in one dashboard
- Including unnecessary information
- Representing data in the wrong type of graph or chart
- Using a cluttered, confusing, or inaccessible design
- Including too many colors
- Including decimals

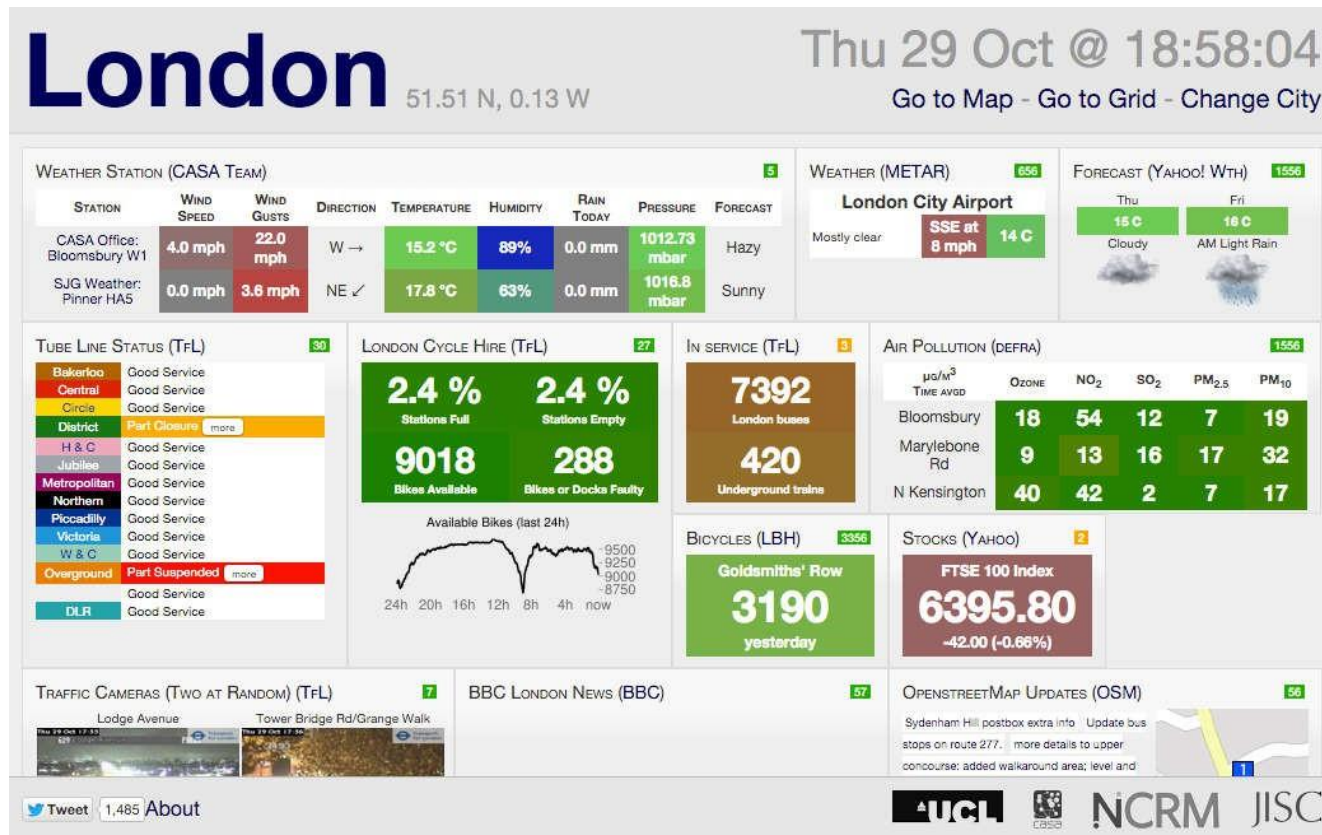
Example: Sales Dashboard



Example: Financial Dashboard



Example: Bad Dashboard Design



Using Tooltips

Tooltips can make the difference between a user loving your visualization and not understanding it.

Tips for improving the basic tooltip

- Use a proper font
- Identify the most important part of the tool tip and make it your title
- Change measure names/values to make them specific and understandable
- Include proper units
- Remove command prompts

Ship Status:	Shipped Late
Customer Name:	Nick Zandusky
Order Date:	7/10/2016
Order ID:	CA-2016-134222
Product Name:	GBC Standard Therm-A-Bind Covers
Ship Mode:	Same Day
Days to Ship Actual:	1

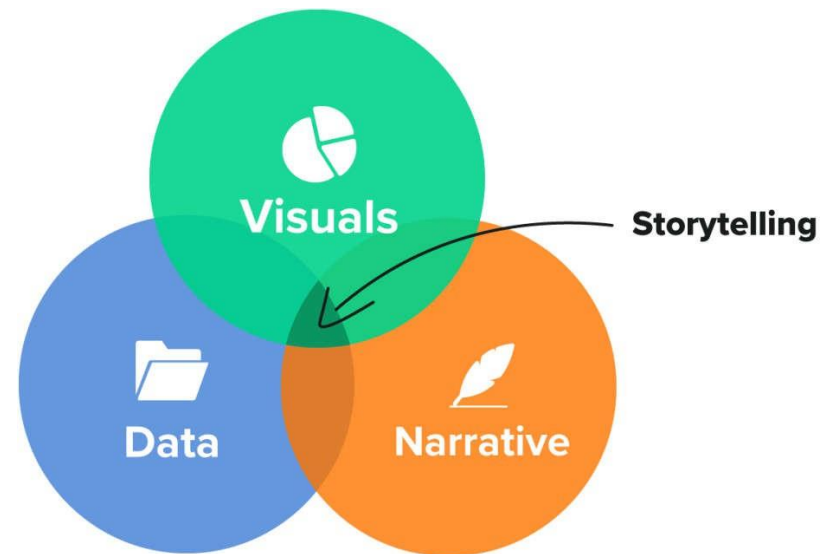
**Default
vs
Modified**

Same Day - Shipped Late	
1 days to ship	
Product Name:	GBC Standard Therm-A-Bind Covers
Customer Name:	Nick Zandusky
Order Date:	7/10/2016
Order ID:	CA-2016-134222

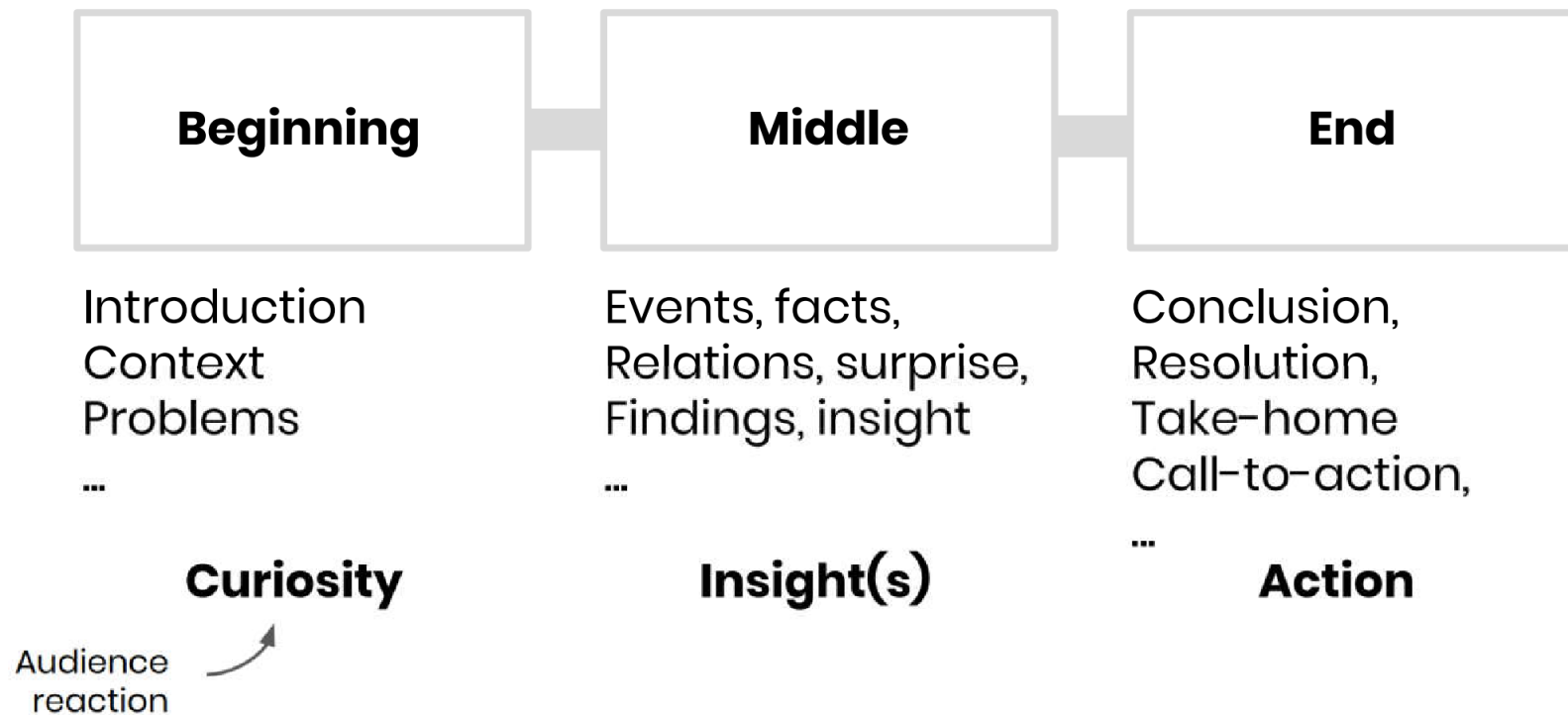
Storytelling

Data Storytelling

- Data visualization expert Stephen Few said, “Data storytelling is a structured approach for communicating data insights, and it involves a combination of three key elements: data, visuals, and narrative”.



Story Structure



The Beginning

- The first thing to do is introduce the plot, building the context for your audience.
- Set up the essential elements of story:
 - the setting
 - main character
 - unresolved state of affairs
 - desired outcome
- Involve your audience, stimulating their interest and answering the questions that are likely on their mind:
 - *Why should I pay attention?*
 - *What is in it for me?*

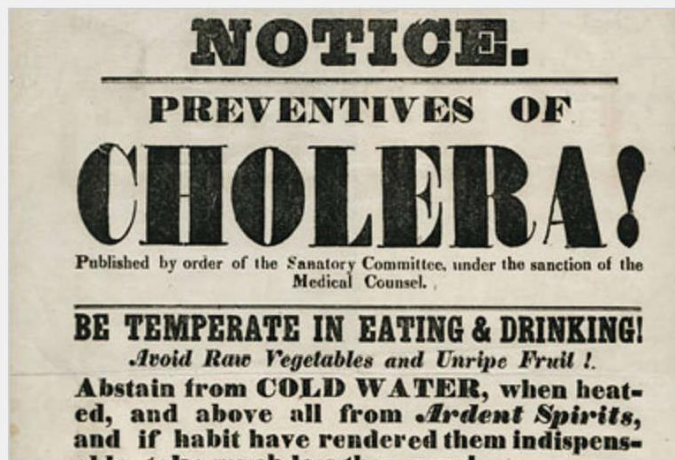
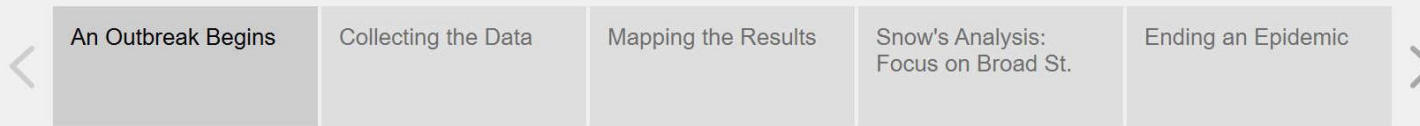
The Middle

- You retain your audience's attention through this part of the story by addressing *how* they can solve the problem you introduced.
 - Further develop the situation or problem by covering relevant background.
 - Incorporate external context or comparison points.
 - Give examples that illustrate the issue.
 - Include data that demonstrates the problem.
 - Articulate what will happen if no action is taken or no change is made.
 - Discuss potential options for addressing the problem.
 - Illustrate the benefits of your recommended solution.
 - Make it clear to your audience why they are in a unique position to make a decision or drive action.

The End

- End the story with a call to action:
 - make it totally clear to your audience what you want them to *do* with the new understanding or knowledge that you've imparted to them.
- One classic way to end a story is to tie it back to the beginning.
 - At the beginning of our story, we set up the plot and introduced the dramatic tension.
 - To wrap up, you can think about recapping this problem and the resulting need for action, reiterating any sense of urgency and sending your audience off ready to act.

London's 1854 Cholera Outbreak: Data Mapping Halts an Epidemic



Victorian medical advice was frequently off the mark by modern standards.

In 1854, a Cholera outbreak swept through the Soho neighborhood of London.

616 people died.

Physician John Snow was skeptical of existing theories of disease transmission, which often blamed "miasmas," or bad air. The germ theory of disease circulation had not yet been outlined.

London's 1854 Cholera Outbreak: Data Mapping Halts an Epidemic

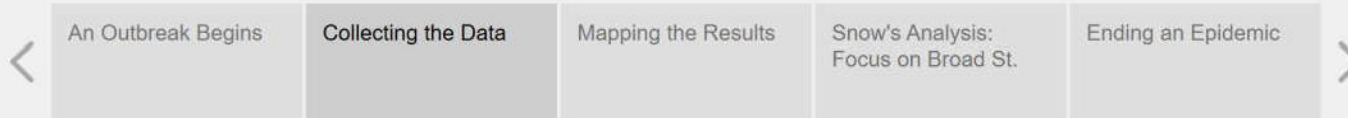


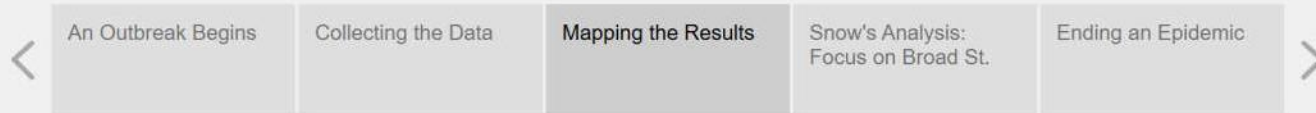
Table 1

Place	No. of Cases	No. of Deaths
1	1	1
2	1	1
3	1	1
4	1	1
5	1	1
6	1	1
7	1	1
8	1	1
9	1	1
10	1	1
11	1	1
12	1	1
13	1	1
14	1	1
15	1	1
16	1	1
17	1	1
18	1	1
19	1	1
20	1	1
21	1	1
22	1	1
23	1	1
24	1	1
25	1	1
26	1	1
27	1	1
28	1	1
29	1	1
30	1	1
31	1	1
32	1	1
33	1	1
34	1	1
35	1	1
36	1	1
37	1	1
38	1	1
39	1	1
40	1	1
41	1	1
42	1	1
43	1	1
44	1	1
45	1	1
46	1	1
47	1	1
48	1	1
49	1	1
50	1	1
51	1	1
52	1	1
53	1	1
54	1	1
55	1	1
56	1	1
57	1	1
58	1	1
59	1	1
60	1	1
61	1	1
62	1	1
63	1	1
64	1	1
65	1	1
66	1	1
67	1	1
68	1	1
69	1	1
70	1	1
71	1	1
72	1	1
73	1	1
74	1	1
75	1	1
76	1	1
77	1	1
78	1	1
79	1	1
80	1	1
81	1	1
82	1	1
83	1	1
84	1	1
85	1	1
86	1	1
87	1	1
88	1	1
89	1	1
90	1	1
91	1	1
92	1	1
93	1	1
94	1	1
95	1	1
96	1	1
97	1	1
98	1	1
99	1	1
100	1	1
Total	600	600

This is Snow's original table showing the chronology of deaths and their total. He notes that not all deaths have recorded addresses; thus, his data set is partially incomplete.

Dr. Snow canvassed the neighborhood. He collected the addresses of those who died, noted the number of deaths at each location, and tabulated the results.

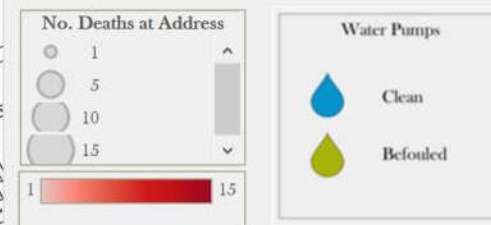
London's 1854 Cholera Outbreak: Data Mapping Halts an Epidemic



Then, Snow mapped these data points onto a map of the Soho neighborhood.

The results were startling.

NB: The map reprinted here is Snow's original. I geocoded the addresses, introduced the number of deaths as a dimension in the dataset, and uploaded the map as a background image. Then, I conformed the dimensions of the map to the appropriate lat/long. Thus, the locations appear as they would have on Snow's map.



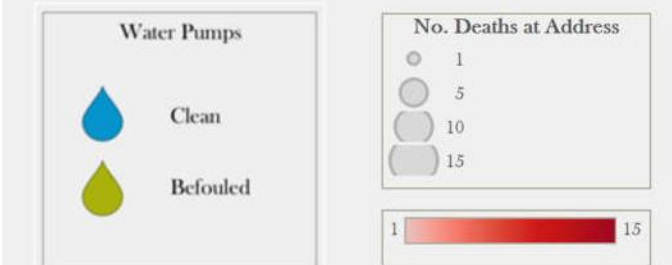
London's 1854 Cholera Outbreak: Data Mapping Halts an Epidemic

←
An Outbreak Begins
Collecting the Data
Mapping the Results
**Snow's Analysis:
Focus on Broad St.**
Ending an Epidemic
→

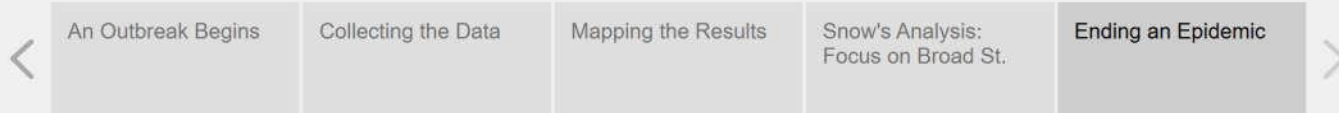
He noted that the bulk of deaths were concentrated in an area that generally used the same municipal water pump: Broad Street.

Snow realized that the Broad Street water pump--and poor water quality--was likely the source of the outbreak.

Armed with this information, he went to city ..



London's 1854 Cholera Outbreak: Data Mapping Halts an Epidemic



Snow advised that the Broad Street water pump be shut down immediately. City authorities found Snow's evidence persuasive, and they removed the handle from the pump.

The outbreak ended shortly thereafter.



John Snow's pioneering data analysis proved that cholera is a water-borne disease, and paved the way for the field of epidemiology.

It is one of the foundational cases of the systematic use of public health information to save lives.

